

KENDRIYA VIDYALAYA SANGATHAN DEHRADUN REGION
BLUE PRINT PREBOARD -I EXAM 2025 – 26
CLASS XII
SUBJECT – MATHEMATICS

Chapter No.	Chapter Name	TYPE OF QUESTIONS						
		MCQ	A-R	V.S.A	S.A	L.A	CASE BASED/ CCT	TOTAL
1	RELATIONS AND FUNCTIONS						1(4)	1(4)
2	TRIGONOMETRIC FUNCTIONS	1(1)	1(1)	1(2)				3(4)
3	MATRICES	3(1)		-	-			3(3)
4	DETERMINANT	2(1)				1(5)		3(7)
5	CONTINUITY AND DIFFERENTIABILITY	2(1)		2(2)	1(3)	-	-	5(9)
6	APPLICATION OF DERIVATIVE	1(1)			1(3)	-	1(4)	3(8)
7	INTEGRATION	2(1)		1(2)		1(5)		4(9)
8	APPLICATION OF INTEGRATION			-	1(3)			1(3)
9.	DIFFERENTIAL EQUATION	1(1)				1(5)		2(6)
10.	VECTORS	3(1)	1(1)	1(2)				5(6)
11	THREE DIMENSIONAL GEOMETRY				1(3)	1(5)		2(8)
12	LINEAR PROGRAMMING	2(1)			1(3)			3(5)
13	PROBABILITY	1(1)			1(3)		1(4)	3(8)
TOTAL		18(18)	2(2)	5(10)	6(18)	4(20)	3(12)	38(80)

KENDRIYA VIDYALAYA SANGATHAN JAMMU REGION(WINTER STATIONS)
CUMULATIVE EXAM 2023-24
CLASS-XI
SUBJECT-MATHEMATICS

TIME ALLOWED:3 HOURS

MAX. MARKS:80

General Instructions:

1. This question paper contains five section. Each section is compulsory
2. Section **A** comprises of **20** MCQ'S of **one mark** each.
3. Section **B** comprises of 5 questions of **2 marks** each.
4. Section **C** comprises of 6 questions of **3 marks** each.
5. Section **D** comprises of 4 questions of **5 marks** each.
6. Section **E** comprises of 3 questions (Case based/CCT) of **4 marks** each.
7. Internal choice is provided in each section. You have to attempt only one of the alternatives in all such questions.

Sr. No .	SECTUIN-A All questions are compulsory. In case of internal choices attempt any one.	Ma rks
1	If $X = \{8^n - 7n - 1 \mid n \in N\}$ and $Y = \{49n - 49 \mid n \in N\}$. Then (a) $X \subset Y$ (b) $Y \subset X$ (c) $X = Y$ (d) $X \cap Y = \emptyset$	1
2	If A and B be two sets in the Universal set, then $(A - B)$ equals: (a) $A \cap B'$ (b) $A' \cap B$ (c) $A \cap B$ (d) <i>none of these</i>	1
3	Let R be the relation in the set N given by $R = \{(a,b): a = b - 2, b \geq 6\}$. Choose the correct answer,, (a) $(2,4) \in R$ (b) $(3,8) \in R$ (c) $(6,8) \in R$ (d) $(8,7) \in R$	1
4	If the set A and B have 2 and 3 elements respectively then the number of subsets of set $(A \times B)$ is: (a) 32 (b) 64 (c) 128 (d) 6	1

5	The domain and range of real function f defined by $f(x) = \frac{4-x}{x-4}$ is given by: (a) Domain = R , Range = $\{-1, 1\}$ (b) Domain = $R - \{1\}$, Range = R (c) Domain = $R - \{4\}$, Range = $\{-1\}$ (d) Domain = $R - \{-4\}$, Range = $\{-1, 1\}$	1
6	If $\tan x = \frac{-4}{3}$ then $\sin x$ is (a) $\frac{-4}{5}$ but not $\frac{4}{5}$ (b) $\frac{-4}{5}$ or $\frac{4}{5}$ (c) $\frac{4}{5}$ but not $\frac{-4}{5}$ (d) none of these	1
7	The value of $\cos 5\pi$ is (a) 0 (b) -1 (c) 1 (d) none of these	1
8	The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \dots \dots \tan 89^\circ$ is equal to (a) 0 (b) ∞ (c) 1 (d) none of these	1
9	The value of i^{-998} is: (a) -1 (b) 1 (c) i (d) $-i$	1
10	The modulus of $3 - 4i$ is: (a) -5 (b) 5 (c) $\sqrt{7}$ (d) $\sqrt{17}$	1
11	If the roots of the equation $x^2 - x + p = 0$ are real, then the value of p is (a) $p \leq 2$ (b) $p \leq 1$ (c) $p \leq \frac{-1}{4}$ (d) $p \leq \frac{1}{4}$	1
12	Which of the following condition does not holds on $\sqrt{ab} = \sqrt{a}\sqrt{b}$ (a) Both a & b are positive (b) Both a & b are negative (c) either of a or b be negative (d) none of these	1
13	If $-5x + 19 \leq -11$, then (a) $x \in (-\infty, -6)$ (b) $x \in (-\infty, 6]$ (c) $x \in (6, \infty)$ (d) $x \in [6, \infty)$	1
14	In how many ways a committee consisting of 3 men and 2 women, can be chosen from 7 men and 5 women? (a) 45 (b) 350 (c) 4200 (d) 230	1
15	The number of signals that can be sent by 6 flags of different colours taking one or more at a time is (a) 63 (b) 21 (c) 720 (d) 1956	1
16	The total number of terms in the expansion of $(x + a)^{51} - (x - a)^{51}$ after simplification is (a) 102 (b) 25 (c) 26 (d) None of these	1
17	The coefficient of y in the expansion of $(\frac{y^2+c}{y})^5$ is (a) $10c$ (b) $10c^2$ (c) $10c^3$ (d) none of these	1
18	In an A.P. the p th term is q and the $(p + q)$ th term is 0. Then the q th term is (a) $-p$ (b) p (c) $p + q$ (d) $p - q$	1

19	Assertion(A): Let $A=\{a,b\}$ and $B=\{x,y\}$. Then number of relations from A to B is 16. Reason(R): If $n(A)=p$ & $n(B)=q$, then number of relations is 2^{pq}. (a) Both assertion and reason are true and reason is the correct explanation of assertion. (b) Both assertion and reason are true but reason is not the correct explanation of assertion. (c) Assertion is true but reason is false. (d) Assertion is false but reason is true.	1
20	Assertion(A): The term independent of x in the expansion of $(x + \frac{1}{x} + 2)^m$ is $\frac{(4m)!}{(2m!)^2}$ Reason(R): The coefficient of x^6 in the expansion of $(1 + x)^n$ is n_{C_6}. (a) Both assertion and reason are true and reason is the correct explanation of assertion. (b) Both assertion and reason are true but reason is not the correct explanation of assertion. (c) Assertion is true but reason is false. (d) Assertion is false but reason is true.	1
	SECTION-B	
21	If $A = \{x: x \text{ is a natural number}\}$, $B = \{x: x \text{ is an even prime number}\}$ then find (i) $A \cap B$ (ii) $A - B$	2
22	If x and y are two different complex numbers with $ y =1$, then $\left \frac{y-x}{1-\overline{x}y} \right $.	2
23	Solve the inequalities: $\frac{(2x-1)}{3} \geq \frac{(3x-2)}{4} - \frac{(2-x)}{5}$.	2
24	Determine n if $2n_{C_3} : n_{C_3} = 11:1$	2
25	Find the number of arrangements of the letters of the word INDEPENDENCE. In how many of these arrangements, (i) do all the vowels always occur together. (ii) do the words begin with I and end with P?	2
	SECTION-C	
26	Let R be a relation from Q to Q defined by $R = \{(a, b): a, b \in Q \text{ and } a - b \in Z\}$. Show that (i) $(a, a) \in R$ for all $a \in Q$ (ii) $(a, b) \in R$ implies that $(b, a) \in R$ (iii) $(a, b) \in R$ and $(b, c) \in R$ implies that $(a, c) \in R$	3
27	In an examination, a question paper consists of 12 questions divided into two parts i.e., Part I and Part II, containing 5 and 7 questions, respectively. A student is required to attempt 8 questions in all, selecting at least 3 from each part. In how many ways can a student select the questions ?	3

28	A manufacturer has 600 litres of a 12% solution of acid. How many litres of a 30% acid solution must be added to it so that acid content in the resulting mixture will be more than 15% but less than 18%?	3
29	Express $(-2 + \frac{1}{3}i)^3$ into a+ib form.	3
30	The sum of two numbers is 6 times their geometric mean, show that numbers are in the ratio $(3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$.	3
31	Using binomial theorem, prove that $6^n - 5n$ always leaves remainder 1 when divided by 25.	3
SECTION-D		
32	(a) Find the sum of $0.6 + 0.66 + 0.666 + \dots$ upto n terms. (3) (b) Find the sum of all numbers between 200 and 400 which are divisible by 7. (2)	5
33	Expand using binomial theorem $(1 + \frac{x}{2} - \frac{2}{x})^4, x \neq 0$.	5
34	(a) Find the modulus and conjugate of $\frac{(3-2i)(2+3i)}{(1+2i)(2-i)}$ (3) (b) Find the multiplicative inverse of $-\sqrt{5} + 3i$ (2)	5
35	(a) Find the value of $\tan \frac{\pi}{8}$ (2) (b) Prove that $\cos 6x = 32\cos^6 x - 48\cos^4 x + 18\cos^2 x - 1$ (3)	5
SECTION-E		
36	CASE BASED-1 Ordered Pairs: The ordered pair of two elements a & b is denoted by (a,b): a is the first element and b is the second element. Two ordered pairs are equal if their corresponding elements are equal i.e (a,b)=(c,d) $\Rightarrow a=c$ & $b=d$. Cartesian Product of Two Sets: For two non-empty sets A & B, the Cartesian product $A \times B$ is the set of all ordered pairs of elements from sets A and B. In symbolic form, it can be written as $A \times B = \{(a,b) : a \in A, b \in B\}$ Base upon the above topics, answer the following questions. (i) If $(a + 2, b - 6) = (-2, 3)$, find the value of a & b. (ii) If $(x + 5, -4) = (3 - y, 2x + y)$, find the value of x & y. (iii) Let A & B be two sets such that $A \times B$ consists of 6 elements. If three elements of $A \times B$ are (1,4), (2,6) and (3,6), then show that $(A \times B) \neq (B \times A)$ OR	1 1 2

	Let A & B be two sets such that $A \times B$ consists of 6 elements. If three elements of $A \times B$ are (1,4), (2,6) and (3,6), then find $A \times B$ & $B \times A$. Also check whether $n(A \times B) = n(B \times A)$	
37	CASE BASED-2 In a class test, a teacher asked the students to consider $x + y = 45^\circ$, where x & y are acute angles. Based on the above information, answer the following questions. (i) Find the value of $(1 + \tan x)(1 + \tan y)$? (ii) Find the value of $\sin(x + y) + \cot(x + y) - \cos(x + y)$ (iii) Find the value of $(\cot x - 1)(\cot y - 1)$? OR Find the value of $\operatorname{cosec}(x + y) - \sec(x + y) - \tan(x + y)$	1 1 2
38	CASE BASED-3 Some students decided to go on a tour. There are total 25 students out of which 10 would like to go on tour. There are 3 friends- Bhavishya, Ankit and Anujesh who decide that either all of them will go or none of them will go the tour. Based on the above information, answer the following questions. (i) In how many ways can the students go on tour, if these three friends will not go? (ii) In how many ways can the students go on tour, if these three friends will go? (iii) In how many ways can the students go on tour? OR In how many ways can the students go on tour if there is no restriction?	4

KENDRIYA VIDYALAYA SANGATHAN JAMMU REGION(WINTER STATIONS)

CUMULATIVE EXAM 2023-24

CLASS-XI

SUBJECT-MATHEMATICS

SUGGESTIVE MARKING SCHEME

1	(a)	1
2	(a)	
3	(c)	
4	(b)	
5	(c)	
6	(b)	

7	(b)	
8	(c)	
9	(a)	
10	(b)	
11	(d)	
12	(b)	
13	(d)	
14	(b)	
15	(d)	
16	(c)	
17	(c)	
18	(b)	
19	(a)	
20	(d)	
	SECTION-B	
21	(i) {2} (ii) {1,3,4,5,.....)	1+1
22	Use $ z ^2 = z\bar{z}$ and ans. 1	1+1/2+ 1/2
23	For L.C.M Seperation of variable & constant $(-\infty, 2)$	0.5 1 0.5
24	Use of combination formula For simplification N=6	0.5 1 0.5

25	By using the concept of permutation (i) 16800 (ii) 12600	1+1
	SECTION-C	
26	Each part one mark	1+1+1
27	Three cases arises: (i) $C(5,3) \times C(7,5)$ (ii) $C(5,4) \times C(7,4)$ (iii) $C(5,5) \times C(7,3)$ Total= 420	1 1 1
28	Let x litres of 30% acid solution Total solution= (x+600)litres 30% of x+12% of 600 >15% of (x+600) and 30% of x+12% of 600 <18% of (x+600) Solution $120 < x < 300$	0.5 1 1 0.5
29	Correct formula simplification	0.5 2.5
30	Correct formula of A.M & G.M Use of compodendo & dividend simplification	1 0.5 1.5
31	Expansion of binomial theorem $(1 + x)^n$ For a=5 Simplification & get solution	1 0.5 1.5
	SECTION-D	

32	(a) Change into G.P Simplification	1.5
		1.5
	(b) finding a & d formula of sum of n terms in A.P	0.5
		0.5
	sum=8729	1
33	Expansion of binomial theorem $(1 + x)^n$ Simplification & get solution	1
	$\frac{16}{x} + \frac{8}{x^2} - \frac{32}{x^3} + \frac{16}{x^4} - 4x + \frac{x^2}{2} + \frac{x^3}{2} + \frac{x^4}{16} - 5$	3+1
34.	(a) simplification	1
	Modulus & conjugate	1+1
	(b)rationalization	1
	Simplification	1
35	(a) formula of $\tan 2x$	0.5
	put $x = \frac{\pi}{8}$ & solution of quadratic equation	1
	value $\sqrt{2} - 1$	0.5
		1
		2
	(b) use of formula $\cos(x+y), \cos 2x, \cos 3x$ simplification	
SECTION-E		
36	(i)a=-4,b=9	1
	(ii)x=-2,y=0	1
	(iii)A={1,2,3} B={4,6} OR	2
	$A \times B = \{(1,4), (1,6), (2,4), (2,6), (3,4), (3,6)\}$, $B \times A = \{(4,1), (6,1), (4,2), (6,2), (4,3), (6,3)\}$	
	Yes both are equal	

37	(i)2	1
	(ii)1	1
	(iii)2 OR -1	2
38	(i)646646	1
	(ii)170544	1
	(iii)817190 or $C(25,10)$	2